FIRE EMERGENCY RESPONSE PLAN

Task 3 - CrewAI Implementation

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Design Overview

5 Crews

Same as those in the original design

- Emergency Services
- Support Crew
- Medical Crew
- Fire Crew
- Reporting Crew

One Single Flow

One flow defines the whole high level process of the Multi Agent System

Two Routers

Determines if there are injured individuals. If so, it activates the Medical Crew.



Changes in the design

Task	Original Design	Final Implementation	Motive
Emergency Crew	**2 tasks, 2 agents & 1 tool**	**2 tasks, 2 agents & 2 tools**	• Added a tool to divide make the report segmentation added
Support Crew	**1 tasks, 1 agents & 1 tool**	**1 tasks, 1 agents & 1 tools**	-
Fire Crew	**3 tasks, 3 agents & 2 tools**	**3 tasks, 3 agents & 3 tools**	• Added a tool to generate the fire_report properly
Medical Crew	**5 tasks, 4 agents & 2 tools**	**3 tasks, 3 agents & 3 tools**	• Merged Paramedic Agent & Hospital agent into a single agent. • Single routing task instead of having 2 routing tasks • Added an ambulance distribution tool
Reporting Crew	**1 tasks, 1 agents & 0 tools**	**1 tasks, 1 agents & 1 tool**	• Added a tool to strucutre the final response plan properly

Cooperation

Crew	Tools	Input	Output	Task
Emergency Crew	**Fire Prioritization Tool, Report Segmentation Tool**	Fire input and fire_medical_report	Proritized_fire_medical_report and segmented fire_medical_report	Prioritizes the different fire incidents and divides the report into a medical and fire reports
Support Crew	**Add Support Tool**	Fire_medical_report	Updated fire_medical_report	If needed, more adds more resources
Fire Crew	**Fire Resource Allocator, Fire Routing, Fire Suppression Plan Tools**	Fire_report	Fire_report_4	Allocates trucks per fire, calculates their routes and adds the steps needed to put the fire out
Medical Crew	**Ambulance Allocator, Ambulance Routing, Ambulance Causality Distributor Tools**	Medical_report	Medical_report_4	Allocates ambulances per fire, calculates the route of each one and finally allocates the injured to each ambulance.
Reporting Crew	**Data Reader Tool**	fire_medical_report, medical_report_4 and fire_report_4	Plan	Spellchecks and edits the final report

Final Tests

Emergency Services Crew

- The Senior Emergency Coordinator agent prioritizes different fire incidents

```
Order of priorization of fires by IDs: [3, 1, 2].Prioritized fire details:
Fire ID: 3, Injured: 3, Type: electrical, Severity: high Distance to nearest hospital: 0.02
Fire ID: 1, Injured: 7, Type: gas, Severity: medium Distance to nearest hospital: 0.01
Fire ID: 2, Injured: 5, Type: ordinary, Severity: low Distance to nearest hospital: 0.01
Number of fires: 3
```

- The Coordinator agent prioritizes different fire incidents and writes the initial report

```
more_resources_needed, described as: True
description, described as: More resources are needed because there are 3 fires, which is a high number relative to the available ambulances (7) and
firetrucks (2). The ratio of firetrucks per fire (0.6666666666666666) is less than 1, indicating that each fire requires at least one additional fir
etruck. Similarly, the ratio of ambulances per fire (2.3333333333333335) is more than 2, indicating that each fire requires at least two additional
ambulances. Additionally, there are a total number of fires of: 3 and Total injured people across all fires: 15.
injured_people, described as: 15
```

Final Tests

Support Crew

- When support is needed, the Resource Allocation Specialist adds the necessary resources.

```
Support has been added successfully. 1 ambulances and 1 firetrucks have been added.
There are 10 firetrucks to handle 3 fires.
There are 11 ambulances to handle 3 fires.
There are 11 ambulances to heal 15 injured people.
```

Final Tests

Fire Crew

- The fire resource allocation specialist allocates the firetrucks

```
The Fire Resource Allocator tool was utilized with the parameters {'high_severity_multiplier': 2.0, 'medium_severity_multiplier': 1.75, 'low_severity_multiplier': 1.0}. T
he high_severity_multiplier value prioritized fires with a higher severity, allocating two firetrucks if possible, while the medium_severity_multiplier and low_severity_m
ultiplier values allocated one and half and one firetruck respectively to medium and low severity fires. This parameter selection impacted the allocation process by ensur
ing that more resources were allocated to high-severity fires. The tool output showed a summary of the allocation of firetrucks in each fire, with a total of 8 fire truck
s distributed among 5 fires. Specifically, fire 1 received 3 firetrucks (high_severity_multiplier), fire 2 received 2 firetrucks (medium_severity_multiplier), fire 3 rece
ived 1 firetruck (low_severity_multiplier), fire 4 received 1 firetruck (low_severity_multiplier), and fire 5 received 1.5 firetrucks (medium_severity_multiplier). The al
location was done to ensure that the number of fire trucks allocated to each fire is at least one, as per the tool's functionality.
```

- The routing expert calculates the routes

```
Routes for Fire 3 with location: (2.483619921963118, 42.19462849095851)
  Firetruck 2 - Route to fire 3:
    Streets: Carrer dels Semalers -> Carrer de la Faixa -> Carrer de la Ferra
 Xiprer -> Carrer dels Amolls -> Carrer del Ventall -> Carrer de la Tartana -
Carrer del Xaboc -> Carrer Pere Màrtir Rossell VI -> Carrer de la Brida -> Ca
 Plaça de la Unió Europea -> Eix Pirinenc
    Total Time: 425.53 seconds
    Total Distance: 3816.91 meters
```

- The Fire Extinguisher Planner makes a suppression plan:

```
**Fire Suppression Plans:**

1. Ordinary High Severity Fire:
   - Step 1: Use a fire extinguisher rated for ordinary fires to smother the flames.
   - Step 2: Use water to cool surrounding objects and prevent re-ignition.
   - Step 3: Evacuate the area and call for backup personnel.

   Routes: Route 1 (Total Time: 10, Distance: 5) and Route 2 (Total Time: 15, Distance: 7)

2. Electrical Medium Severity Fire:
   - Step 1: Use a fire extinguisher rated for electrical fires to prevent re-ignition.
   - Step 2: Disconnect the power source if possible.
   - Step 3: Evacuate the area and call for backup personnel.

   Routes: Route 3 (Total Time: 12, Distance: 6) and Route 4 (Total Time: 18, Distance: 8)
```

Final Tests

Medical Crew

- The medical service coordinator allocates the ambulances

```
The tool allocated 7 ambulances in total for fire ID 1, with a medium severity. This is based on the number of minor (2) and serious (5) injuries. T
here are 3 transfer ambulances and 4 on-site ambulances.
```

- The routing expert calculates the routes

```
* Total Number of Ambulances: 7
* Hospitals assigned to ambulances: Hospital A, Hospital B
* Injuries: Minor (2), Serious (5)
* Transfer Ambulances: 3
* On-site Ambulances: 4

**Ambulance Routes:**
```

```
1. **Ambulance 1**
        * Route from Fire Location to Hospital d'Olot i comarcal de la Garrotxa (ID: 1):
                + Streets: Camí dels Tossols -> Carrer de Cadis -> Carrer de Ribes de Freser -> Carrer de la Coma -> Plaça de Vista Alegre -> Camí d
e la Teuleria -> Carretera Santa Coloma -> Passeig de Barcelona -> Carrer del Pare Roca -> Carretera de Sant Joan de les Abadesses -> Avinguda dels
Països Catalans
                + Route Time: 456.63 seconds
                + Route Distance: 4801.75 meters
```

Final Tests

Reporting Crew

- The tool creates a written report

```
 Initial Status

 There are 3 fires in the area.
Fire 1 is located at {'x': 2.4639846209913387, 'y': 42.17738048019025}. It is a/an gas fire, and it is of medium
severity, with 2 injured.
There are 2 persons with minor injuries and 5 with serious injuries.
Fire 3 is located at {'x': 2.483619921963118, 'y': 42.19462849095851}. It is a/an electrical fire, and it is of
high severity, with 2 injured.
There are 1 persons with minor injuries and 2 with serious injuries.
Fire 2 is located at {'x': 2.4935978583909413, 'y': 42.1738600307819}. It is a/an ordinary fire, and it is of low
severity, with 2 injured.
There are 1 persons with minor injuries and 4 with serious injuries.
```

- The agent often fails to spell check it and offers a summarized version of the firefighter plans

```
## Final Answer:
The details provided in the prompt are about various incidents involving fire safety, including actions taken by firefighters. The main points from these incidents can be
 summarized as follows:
```

Code Structure



- One Folder per Crew
- One Single Folder
- One folder for the schemas. A different file for each schama
- One folder for the schemas. A different file for each schama

Thank you!

